

Writing Large Numbers in Scientific NotationQuestion Bank 3.08 **ANSWER KEY**

Show your work. Circle your final answer.

For questions 1 through 4, write each number in scientific notation.

1. 9,000

$$9 \times 10^3$$

2. 4,500,000

$$4.5 \times 10^6$$

3. 310,000,000

$$3.1 \times 10^8$$

4. 7,020,000,000

$$7.02 \times 10^9$$

The zero between the 7 and the 2 stays in the coefficient.

For questions 5 and 6, write each number in standard form.

5. 6.3×10^4

$$63,000$$

6. 1.05×10^8

$$105,000,000$$

Real quantities.

7. Write each quantity in scientific notation.

- a. There are about 1,670,000,000,000,000,000 molecules in a drop of water.

$$1.67 \times 10^{21} \text{ molecules}$$

- b. The average distance from Earth to the Sun is about 93,000,000 miles.

$$9.3 \times 10^7 \text{ miles}$$

- c. A computer displays the number of stars in a galaxy as **4E11**. Write it in standard form.

$$400,000,000,000$$

E11 means $\times 10^{11}$.

8. Complete each comparison.

- a. How many times larger is 8.1 than 5.4?

1.5

- b. How many times larger is 10^8 than 10^7 ?

10

- c. How many times larger is 8.1×10^8 than 5.4×10^7 ?

$1.5 \times 10 = 15$ times larger

9. Answer each question.

- a. A blue whale has a mass of about 4×10^5 kg. An elephant has a mass of about 1×10^4 kg. How many times larger is the whale's mass?

40 times

$4 \div 1 = 4$ and $10^5 \div 10^4 = 10$.

- b. Florida has an area of about 6.5×10^4 square miles. Lake Okeechobee has an area of about 7.3×10^2 square miles. About how many times larger is Florida? Round to the nearest whole number.

about 89 times

$6.5 \div 7.3 \approx 0.89$ and $10^4 \div 10^2 = 100$; $0.89 \times 100 \approx 89$. Directly: $65,000 \div 730 \approx 89.04$.

10. Which of the following is greater? Explain how you know.

- a. 9.5×10^5 or 1.2×10^6

1.2×10^6 — 10^6 is greater than 10^5 , so the powers decide; $1,200,000 > 950,000$.

- b. 3.8×10^7 or 3.09×10^7

3.8×10^7 — same power, so compare coefficients: $3.8 > 3.09$.

11. Which is 52,000,000 written in scientific notation?

● **A. 5.2×10^7**

○ B. 52×10^6

○ C. 5.2×10^6

○ D. 5.2×10^8

A

12. Select ALL of the expressions that are equal in value to 3,400,000.

- ☒ A. 3.4×10^6
- ☒ B. 34×10^5
- ☒ C. 340×10^4
- ☒ D. 0.34×10^7
- ☐ E. 3.4×10^7
- ☐ F. 3.4×10^5

A, B, C and D

Only A is in scientific notation; B, C and D are equal in value but their coefficients are not in $[1, 10)$.

13. In question 12, only one of the four equal expressions is written in scientific notation. Which one is it, and why are the other three not?

3.4×10^6 . In scientific notation the coefficient must be at least 1 and less than 10; 34, 340 and 0.34 are not.